

KULSARY, KazakhstanWMO: **357150**Lat: **46.967N**Lon: **54.017E**Elev: **18**StdP: **101.11**Time Zone: **5.00 (E05)**Period: **82-94**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.4	-21.2	-26.3	0.3	-22.7	-24.3	0.4	-20.5	20.4	-7.6	20.1	-7.4	3.6	50	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	38.0	19.4	35.9	19.1	34.0	18.4	21.7	31.4	20.8	30.7	20.0	29.6	5.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.9	24.4	17.8	12.8	24.9	16.5	11.8	24.4	63.4	31.0	60.2	30.2	57.5	29.4	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.1	14.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-7.6	-9.6	-1.7	11.5	18.0	24.6	26.7	24.7	17.3	8.5	-1.2	-5.9	(6)
(7)		DBStd	13.89	5.07	6.62	6.54	4.89	4.24	3.75	3.47	3.48	4.43	4.32	6.12	5.58	(7)
(8)		HDD10.0	2411	547	549	364	39	1	0	0	0	3	78	337	492	(8)
(9)		HDD18.3	4196	805	783	622	212	59	1	1	2	71	304	587	750	(9)
(10)		CDD10.0	2002	0	0	1	84	249	439	519	457	221	33	0	0	(10)
(11)		CDD18.3	746	0	0	0	6	49	190	261	201	39	0	0	0	(11)
(12)		CDH23.3	8936	0	0	0	63	616	2259	3317	2229	446	7	0	0	(12)
(13)		CDH26.7	4349	0	0	0	8	218	1136	1783	1050	153	0	0	0	(13)
(14)	Wind	WSAvg	5.8	6.1	7.0	6.7	6.3	5.6	5.3	5.0	5.3	5.3	5.6	6.0	(14)	
(15)	Precipitation	PrecAvg	155	11	10	12	17	16	24	13	5	6	12	14	15	(15)
(16)		PrecMax	252	26	19	43	46	31	71	39	28	16	37	26	32	(16)
(17)		PrecMin	85	1	3	0	1	2	1	0	0	0	2	2	4	(17)
(18)		PrecStd	49	6	4	11	14	9	22	10	7	5	11	7	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	4.2	15.1	27.7	33.4	39.7	40.7	37.7	33.0	24.5	11.4	5.4	(19)
(20)			MCWB	1.5	2.9	8.7	14.2	16.2	19.8	20.1	19.9	16.3	12.9	8.4	4.0	(20)
(21)		2%	DB	1.2	1.2	12.1	25.4	31.1	37.1	38.9	35.6	30.6	21.6	9.5	3.5	(21)
(22)			MCWB	0.1	0.4	7.1	12.7	15.8	18.9	20.2	19.4	15.2	12.0	6.8	2.6	(22)
(23)		5%	DB	0.6	0.4	9.7	22.8	29.1	35.0	36.7	33.8	28.3	18.5	7.7	2.0	(23)
(24)			MCWB	-0.3	-0.5	5.7	12.0	15.1	18.6	19.4	18.5	14.8	10.2	5.3	1.2	(24)
(25)		10%	DB	-0.2	-0.6	6.9	20.3	27.0	32.9	34.7	32.0	25.9	15.7	6.1	0.9	(25)
(26)			MCWB	-0.9	-1.4	4.2	11.0	14.2	17.6	19.1	18.0	14.4	9.0	4.3	0.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	3.0	9.7	16.0	19.3	22.2	23.5	22.6	18.5	14.2	9.2	4.6	(27)
(28)			MCDWB	2.4	3.6	14.3	23.6	26.4	31.7	33.4	31.3	27.2	22.3	10.9	5.3	(28)
(29)		2%	WB	0.5	0.7	7.4	14.3	17.6	20.8	22.1	21.1	17.0	12.6	7.3	2.8	(29)
(30)			MCDWB	1.0	1.1	11.0	21.4	27.2	29.8	32.2	30.4	26.7	18.7	9.0	3.5	(30)
(31)		5%	WB	-0.2	-0.3	5.8	13.2	16.4	19.9	21.2	20.2	15.7	11.1	5.8	1.5	(31)
(32)			MCDWB	0.4	0.3	9.3	20.6	26.0	29.7	31.8	29.5	25.8	17.1	7.2	1.9	(32)
(33)		10%	WB	-1.0	-1.3	4.2	11.8	15.3	19.1	20.4	19.4	14.6	9.7	4.2	0.3	(33)
(34)			MCDWB	-0.2	-0.6	6.9	19.1	24.5	28.8	30.6	28.7	24.2	14.9	5.9	0.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.1	8.2	8.2	11.6	12.2	12.1	11.9	11.8	12.3	11.1	6.9	6.2	(35)
(36)			MCDBR	6.8	6.2	11.3	14.5	15.4	14.9	14.7	14.4	16.0	15.6	8.5	5.5	(36)
(37)			MCWBR	6.0	5.6	7.1	6.7	6.7	5.6	5.9	5.8	7.5	8.7	6.1	5.0	(37)
(38)		5% WB	MCDBR	6.2	5.6	10.3	12.6	13.2	12.3	11.9	12.0	14.4	13.5	6.9	4.9	(38)
(39)			MCWBR	5.8	5.2	6.8	6.7	7.0	5.9	5.4	5.7	7.8	8.4	6.2	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.347	0.371	0.434	0.485	0.474	0.458	0.451	0.449	0.402	0.396	0.370	0.358	(40)	
(41)		taud	2.051	2.009	1.980	1.909	1.998	2.089	2.133	2.110	2.192	2.133	2.142	2.064	(41)	
(42)		Ebn at Noon	686	759	768	769	797	813	813	794	799	726	660	617	(42)	
(43)		Edh at Noon	98	126	152	180	170	156	148	145	122	111	90	87	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.13	2.10	3.43	4.95	6.25	6.98	6.69	5.93	4.53	2.78	1.49	0.90	(44)	
(45)		RadStd	0.22	0.36	0.36	0.41	0.44	0.48	0.33	0.25	0.28	0.24	0.17	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+1.33	N/A	-0.73	N/A	N/A	+231	+155

Nomenclature: See separate page